WORKED SOLUTION

STEP II 2012, Question 9

Mechanics • Projectile motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

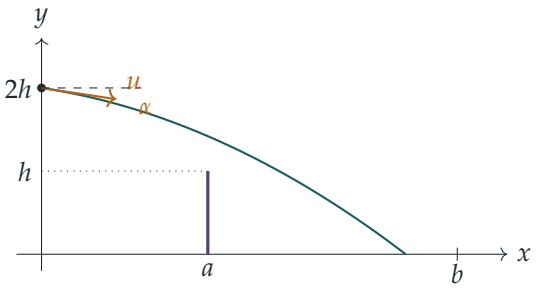
step.maths.org/questions/12-s2-q9

The key idea

Write the height of the ball as a function of its horizontal distance x from the point of projection: the trajectory equation. "Passes over the net" is then one inequality at $x = a$, and "lands before b" is an inequality on the time of flight. The final part works because both conditions turn out to be bounds on the same quantity $1/u^2$, one from above and one from below, so the lower bound must be smaller than the upper one, and u drops out.

Solution

We model the ball as a particle moving freely under gravity, ignoring air resistance and spin, with $0 \leq \alpha < \frac{1}{2}\pi$. Take the origin on the ground directly below the point of projection, x horizontal towards the net and y vertically upwards.



Passing over the net

The initial velocity has components $u\cos\alpha$ horizontally and $u\sin\alpha$ *downwards*. After time t,

$$x = ut\cos\alpha, \qquad y = 2h - ut\sin\alpha - \tfrac{1}{2}gt^2.$$

Eliminating $t = x/(u\cos\alpha)$ gives the trajectory

$$y = 2h - x\tan\alpha - \frac{gx^2\sec^2\alpha}{2u^2}.$$

The horizontal velocity is constant and positive, so the ball reaches $x = a$ exactly once, and it clears the net when $y > h$ there:

$$2h - a\tan\alpha - \frac{ga^2\sec^2\alpha}{2u^2} > h \iff \frac{ga^2\sec^2\alpha}{2u^2} < h - a\tan\alpha.$$

Multiplying by the positive quantity $2/(ga^2\sec^2\alpha)$,

$$\frac{1}{u^2} < \frac{2(h - a\tan\alpha)}{ga^2\sec^2\alpha},$$

as required.

Landing before b

The ball lands when it has fallen $2h$, that is when $\frac{1}{2}gt^2 + ut\sin\alpha - 2h = 0$. This quadratic in t has roots of opposite sign (their product is $-4h/g < 0$), and the landing time is the positive one:

$$T = \frac{-u\sin\alpha + \sqrt{u^2\sin^2\alpha + 4gh}}{g}.$$

It lands before horizontal distance b exactly when $uT\cos\alpha < b$, i.e. $T < \dfrac{b}{u\cos\alpha}$ (as $u\cos\alpha > 0$). Multiplying by g and adding $u\sin\alpha$,

$$\sqrt{u^2\sin^2\alpha + 4gh} < \frac{bg}{u\cos\alpha} + u\sin\alpha,$$

which is the second result.

The final inequality

Ask yourself How can two conditions involving u produce a condition with no u in it?

The first result is an upper bound on $1/u^2$. If we can turn the second into a *lower* bound on $1/u^2$, then the lower bound must be less than the upper bound, whatever u is.

Both sides of the second inequality are positive (the right-hand side because $\sin\alpha \geq 0$ and $\cos\alpha > 0$), so squaring preserves it:

$$u^2\sin^2\alpha + 4gh < \frac{b^2g^2}{u^2\cos^2\alpha} + 2bg\tan\alpha + u^2\sin^2\alpha.$$

The $u^2\sin^2\alpha$ terms cancel, leaving $2g(2h - b\tan\alpha) < b^2g^2\sec^2\alpha/u^2$, that is

$$\frac{2(2h - b\tan\alpha)}{gb^2\sec^2\alpha} < \frac{1}{u^2}.$$

Combining with the first result,

$$\frac{2(2h - b\tan\alpha)}{gb^2\sec^2\alpha} < \frac{1}{u^2} < \frac{2(h - a\tan\alpha)}{ga^2\sec^2\alpha}.$$

Cancelling the positive factor $2/(g\sec^2\alpha)$ and multiplying through by $a^2b^2 > 0$,

$$\begin{aligned} & a^2(2h - b\tan\alpha) < b^2(h - a\tan\alpha) \\ \iff \quad & (ab^2 - a^2b)\tan\alpha < h(b^2 - 2a^2) \\ \iff \quad & ab(b - a)\tan\alpha < h(b^2 - 2a^2). \end{aligned}$$

Ask yourself Can we divide by $b - a$?

Yes, and the sign matters. The ball is still above height $h > 0$ at $x = a$, so it lands at some distance greater than a; and it lands before b. Hence $b > a$, so $ab(b - a) > 0$ and dividing keeps the inequality the right way round:

$$\tan\alpha < \frac{h(b^2 - 2a^2)}{ab(b - a)}.$$

Checks

- Units: $a\tan\alpha$ is a length like h, and ga^2/u^2 is a length too, so each bound on $1/u^2$ has units $\mathrm{s}^2\,\mathrm{m}^{-2}$, as it should.
- With $\alpha = 0$ the final result says $b^2 > 2a^2$. Indeed, a ball fired horizontally has fallen less than h by $x = a$; since the drop is proportional to x^2, it has fallen less than $2h$ by $x = \sqrt{2}\,a$, so it lands beyond $\sqrt{2}\,a$ and $b > \sqrt{2}\,a$.
- Numerically, with $g = 9.8$, $u = 12$, $\alpha = 0.2$, $h = 1.5$, stepping the motion gives a landing distance of 6.7759, agreeing with $uT\cos\alpha$. In 20 000 random cases the first two inequalities matched "clears the net" and "lands before b" exactly, and the final inequality held whenever both did.

Common slips

- Taking the vertical component of the initial velocity as $+u\sin\alpha$: the ball is projected *below* the horizontal.
- Squaring an inequality without checking that both sides are positive.
- Dividing by $b - a$ without saying why it is positive.

Beyond STEP

The final part is a small instance of eliminating a parameter between inequalities: a system $L < 1/u^2 < U$, with L and U free of u, has a solution exactly when $L < U$.

The same idea, done systematically for many linear inequalities, is Fourier–Motzkin elimination, one of the foundations of linear programming. For the mechanics, including projectiles with air resistance where the trajectory is no longer a parabola, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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